

Assignment 1



The University of Faisalabad



Topic of Assignment

Mental Models in AI



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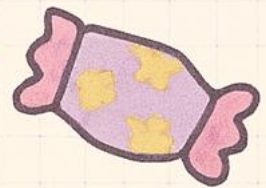
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Mental Models in Understanding Developer Thinking



1. Mental Models: The Psychological Backbone



Mental models are not just thoughts—they are frameworks we use to:

- Understand systems.
- Predict outcomes.
- Explain causality.
- Make decisions in uncertain environments.



In AI, this is extremely important—developers deal with systems that may not always behave as expected (like deep learning).



2. Developer Thinking in AI: A Cognitive Task



AI developers constantly:

- Interpret abstract data.
- Design models based on incomplete information.
- Use intuition (backed by experience) to tune models.



Psychological Relevance:

- Involves working memory, long-term memory, attention, and problem-solving skills.



Mental Models in AI

- Developers engage in situated cognition – thinking based on real-world contexts (like user behavior, business needs).

3. Types of Mental Models Developers Use



Type	Example
Conceptual Models	Understanding how AI works in theory.
Operational Models	"If I change this parameter, accuracy will go up."
User Models	"Users won't trust it unless I explain the decision."
Debugging Models	Used when fixing problems.

4. Real-World Examples

Example 1: ML Model Debugging

A developer expects a model to improve with more data.

It doesn't → cognitive dissonance occurs.

They revise their mental model to realize data quality matters more than quantity.

Example 2: Explainable AI (XAI)

Developers must "teach" users how a model



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5. Misaligned Mental Models

- When mental models are incorrect, or incomplete, this can lead to:
- Wrong design decisions.
- Model bias or failure.
- Difficulty in debugging.
- Communication issues with teammates or stakeholders.



Psychological Tie-In:

Developers must regularly engage in metacognitive regulation—thinking about their own thinking.



6. Mental Models & AI Education

Studies show students new to AI often form faulty mental models like:

- "The computer thinks like a human."
- "More layers = better model" (not always true).





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UNDERSTANDING DEVELOPER THINKING

Teachers now focus on mental model correction to help students reason better.



How Mental Models Grow

(Cognitive Development in AI Learning)

Stages:

- Novice: Relies on surface features (e.g., code syntax).
- Intermediate: Starts recognize patterns (e.g., hyperparameters) -
- Expert: Has layered mental models - can predict behavior under complex scenarios.



Inspired by Bloom's Taxonomy: Developers go from remembering to creating.



8. Research & Insights

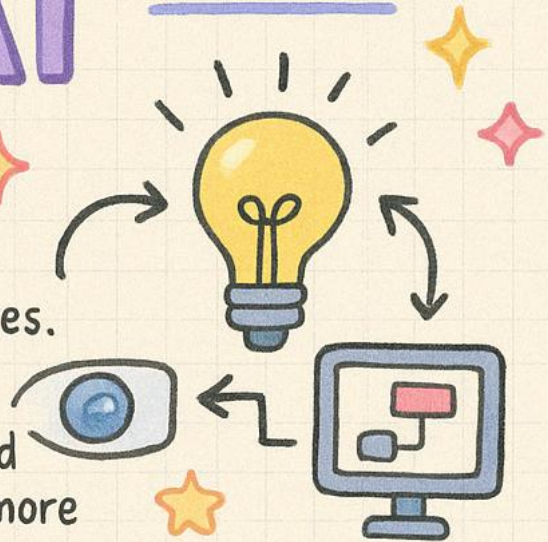
2019 - Kulesza et al.: Found that developers who had better mental models of AI systems were faster and more accurate in debugging.

2022 - Google Research: Proposed tools that "exter-

Mental Models in AI

2022 - Google Research:
Proposed tools that
"externalize" mental models
through diagrams and interfaces.

2023 - MIT Cognitive AI Lab:
Showed that teams with aligned
mental models built safer and more



9. Strategies to Improve Developer Mental Models

👁️ Visualization: Helps convert abstract ideas into concrete images.

★ Reflection: Regular journaling or documentation of thought processes,

• Collaboration: Shared mental models across teams = better results.

Explainability tools: SHAP, LIME, TensorBoard, etc., make models more understandable.

10. Conclusion

Understanding the psychology of how developers think helps not only in designing better AI, but also in creating systems that are more aligned with human needs. Mental models are the invisible tools of every AI developer, shaping their decisions



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Conclusion

Mental models are the invisible tools of every AI developer, shaping their decisions, guiding their experiments, and defining their success.

